

# Optimal block size for variance estimation by a spatial block bootstrap method

running title: bootstrap block size

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## Abstract

This paper considers the block selection problem for a block bootstrap variance estimator applied to spatial data on a regular grid. We develop precise formulae for the optimal block sizes that minimize the mean squared error of the bootstrap variance estimator. We then describe practical methods for estimating these spatial block sizes and prove the consistency of a block selection method by Hall, Horowitz and Jing (1995), originally introduced for time series. The spatial block bootstrap method is illustrated through data examples and its performance is investigated through several simulation studies.

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## 1 Introduction

In recent years, different versions of block bootstrap methods have been proposed for spatial data. As in the time series case, the accuracy of a block bootstrap estimator of a population parameter critically depends on the block size employed. Although this is a very important problem, there seems to be little information available in the literature about the (theoretical) optimal block sizes for estimating a given parameter with a spatial block bootstrap. In this paper, we investigate the problem of determining optimal block sizes for variance estimation by a spatial block bootstrap method.

For times series data, expressions for theoretical optimal block sizes with different block bootstrap methods are known (Künsch, 1989, Hall *et al.*, 1995, Lahiri, 1999). In the time series case, the optimal block sizes depend on the blocking mechanism (i.e., overlapping/non-overlapping blocks) and the covariance structure of the process. In comparison, the optimal

block sizes in the spatial case are determined by the blocking mechanism, the covariance structure of the spatial process, and the dimension of the spatial sampling region. For a large class of “smooth function model” spatial statistics, the main results of this paper give expansions for the mean squared error (MSE) of block bootstrap variance estimators based on overlapping and non-overlapping versions of the spatial block bootstrap method. Using these MSE expansions, we give explicit expressions for the optimal bootstrap block sizes. It turns out that, due to the mechanics of the bootstrap, the optimal blocks in the spatial case are a natural extension of the time series case under identical conditions on the smooth function model. This result is surprising compared to spatial subsampling variance estimators, studied by Politis and Romano (1994a), Sherman and Carlstein (1994) and Sherman (1996). With subsampling, the formulae for optimal blocks may differ largely between the spatial/temporal settings and also require much smoother functions for time series data than for spatial data (Nordman and Lahiri, 2004).

We then develop data-based methods for estimating the optimal block size and study their performance through simulation studies. In particular, we contribute the first proof of consistency (to our knowledge) for a general block estimation technique of Hall *et al.* (1995), which has been applied by Hall and Jing (1996), Politis and Sherman (2001) and Nordman and Lahiri (2004) among others; the result applies to both time series and spatial block bootstraps. We also illustrate the proposed methodology with two data examples, which investigate spatial patterns in longleaf pine trees (Cressie, 1993) and a cancer mortality map of the United States (Sherman and Carlstein, 1994).

The rest of the paper is organized as follows. We conclude this section with a brief literature review. In Section 2, we describe the spatial sampling framework, the variance estimation problem, and the spatial block bootstrap methods for variance estimation. In Section 3, we state the assumptions and the main results on optimal block sizes. Empirical methods for selecting the optimal block sizes are discussed in Section 4 while Section 5 summarizes a simulation study of the bootstrap estimators with empirical block choices. Section 6 describes data examples of the bootstrap method. Proofs of all technical results are given in the Appendix.

Block bootstrap methods for time series data and spatial data have been put forward by Hall (1985), Künsch (1989), Liu and Singh (1992), Politis and Romano (1993, 1994b), Politis *et al.* (1999), Zhu and Lahiri (2007) among others. In contrast to the spatial setting, considerable research has focused on optimal block sizes for time series block bootstrap methods including work by Künsch (1989), Hall *et al.* (1995), Lahiri (1996), Bühlmann

and Künsch (1999), Lahiri *et al.* (2006), and Politis and White (2004). Nordman and Lahiri (2004) derive expressions for optimal block sizes for a class of spatial subsampling methods. See Lahiri (2003a, Chapter 12) for more on bootstrap methods for spatial data.

## 2 Block bootstrap variance estimators

### 2.1 Variance estimation problem with spatial data

We assume that the available spatial data are collected from a spatial sampling region  $R_n \subset \mathbb{R}^d$  as follows. For a fixed vector  $\mathbf{t} \in [-1/2, 1/2)^d$ , identify the  $\mathbf{t}$ -translated integer lattice as  $\mathbf{Z}^d \equiv \mathbf{t} + \mathbb{Z}^d$ . Suppose that a  $p$ -dimensional, stationary weakly dependent spatial process  $\{Z(\mathbf{s}) : \mathbf{s} \in \mathbf{Z}^d\}$  is observed at those locations  $\mathcal{S}_n \equiv \{\mathbf{s}_1, \dots, \mathbf{s}_{N_n}\}$  of the lattice grid  $\mathbf{Z}^d$  that lie inside  $R_n$ , i.e., the data are  $\mathcal{Z}_n = \{Z(\mathbf{s}) : \mathbf{s} \in \mathcal{S}_n\}$  for  $\mathcal{S}_n = R_n \cap \mathbf{Z}^d$ . For simplicity, we let  $N \equiv N_n$  denote the spatial sample size or the number of sites in  $R_n$ .

To allow a wide variety of sampling region shapes, we suppose that  $R_n$  is obtained by inflating a prototype set  $R_0$  by a constant  $\lambda_n$ :

$$R_n = \lambda_n R_0, \tag{1}$$

where  $R_0 \subset (-1/2, 1/2]^d$  is a Borel subset containing an open neighborhood of the origin and  $\{\lambda_n\}_{n \geq 1}$  is a positive sequence of scaling factors such that  $\lambda_n \uparrow \infty$  as  $n \rightarrow \infty$ . Because  $R_0$  contains the origin, the shape of the sampling region  $R_n$  is preserved for different values of  $n$  as the region grows in an “increasing domain asymptotic” framework as termed by Cressie (1993). Sherman and Carlstein (1994), Sherman (1996) and Nordman and Lahiri (2004) use a comparable sampling framework. Observations indexed by  $\mathbf{Z}^d \equiv \mathbf{t} + \mathbb{Z}^d$ , rather than the integers  $\mathbb{Z}^d$ , entail that the “center” of  $R_n$  at the origin need not be a sampling site.

We study the spatial block bootstrap method of variance estimation for a large class of statistics that are functions of sample means. Let  $H : \mathbb{R}^p \rightarrow \mathbb{R}$  be a smooth function and  $\bar{Z}_N = N^{-1} \sum_{i=1}^N Z(\mathbf{s}_i)$  denote the sample mean of the  $N$  sites within  $R_n$ . Suppose that a relevant statistic  $\hat{\theta}_n$  can be represented as a function of the sample mean  $\hat{\theta}_n = H(\bar{Z}_N)$  and estimates the population parameter  $\theta = H(\mu)$  involving the mean  $\mathbb{E}Z(\mathbf{t}) = \mu \in \mathbb{R}^p$  of the random field. Hall (1992) refers to this parameter/estimator framework as the “smooth function” model which permits a wide range of spatial estimators with suitable transformations of the  $Z(\mathbf{s})$ ’s, including means, differences and ratios of sample moments (e.g., spatial variograms), and other spatial test statistics.

Using a spatial block bootstrap, we wish to estimate the variance of the scaled statistic  $\sqrt{N}\hat{\theta}_n$ , say  $\sigma_n^2 = N\text{Var}(\hat{\theta}_n)$ . We next describe two bootstrap variance estimators of  $\sigma_n^2$ .

## 2.2 Block bootstrap methods

The spatial bootstrap aims to generate bootstrap renditions of spatial data through the same recipe used in the moving block bootstrap (MBB) for time series (Künsch, 1989, Liu and Singh, 1992). Recall that the MBB produces a bootstrap reconstruction of a length  $n$  time series through a sequence of block resampling steps as follows.

1. Partition the time series into  $k = \lfloor n/b \rfloor$  consecutive observational blocks of length  $b$ .
2. Develop some collection of length  $b$  observational blocks from the original time series. Two possibilities for block collections are either the (non-overlapping) blocks in Step 1 or the size  $n - b + 1$  set of all possible (potentially overlapping) blocks of length  $b$ .
3. For each data block in Step 1, create a bootstrap rendition of this block by independently resampling a length  $b$  block of observations from the collection in Step 2.
4. Paste the  $k$  resampled blocks together to form a bootstrap time series of length  $kb$ .

To re-create spatial data, the spatial block bootstrap involves the same essential steps as the MBB, namely dividing the sampling region  $R_n$  into spatial data blocks, creating a bootstrap version of each data block through block resampling, and concatenating these resampled blocks together into a bootstrap spatial sample. Details are described next, where the time series bootstrap is modified to accommodate the spatial data structure.

Let  $\{b_n\}_{n \geq 1}$  be a sequence of positive integers to define the  $d$ -dimensional spatial blocks as  $\mathcal{B}_n(\mathbf{i}) \equiv \mathbf{i} + b_n\mathcal{U}$ ,  $\mathbf{i} \in \mathbb{Z}^d$ , using the unit cube  $\mathcal{U} = (0, 1]^d$ . We suppose  $b_n^{-1} + b_n/\lambda_n \rightarrow 0$  as  $n \rightarrow \infty$  to keep the blocks  $\mathcal{B}_n(\cdot)$  small relative to the size of the sampling region  $R_n$  in (1).

To implement the block bootstrap, the sampling region  $R_n$  is first divided into disjoint cubes or blocks of spatial observations. To this end, let  $\mathcal{K}_n = \{\mathbf{k} \in \mathbb{Z}^d : \mathcal{B}_n(b_n\mathbf{k}) \subset R_n\}$  represent the index set of all complete cubes  $\mathcal{B}_n(b_n\mathbf{k}) = b_n(\mathbf{k} + \mathcal{U})$  lying inside  $R_n$ ; Figure 1(a) illustrates a partition of  $R_n$  by cubes. For any  $A \subset \mathbb{R}^d$ , let  $\mathcal{Z}_n(A) = \{Z(\mathbf{s}) : \mathbf{s} \in A \cap \mathcal{S}_n\}$  denote the set of all observations corresponding to the sampling sites lying in the set  $A$ . We define a bootstrap version of  $\mathcal{Z}_n(R_n)$  by putting together bootstrap replicates of the process  $Z(\cdot)$  on each block in the partition of  $R_n$ , given by

$$R_n(\mathbf{k}) \equiv R_n \cap \mathcal{B}_n(b_n\mathbf{k}), \quad \mathbf{k} \in \mathcal{K}_n. \quad (2)$$

That is, we consider one block subregion  $R_n(\mathbf{k})$ ,  $\mathbf{k} \in \mathcal{K}_n$ , at a time and create a bootstrap rendition of the data  $\mathcal{Z}_n(R_n(\mathbf{k}))$  in that block by resampling from a suitable collection of blocks in  $R_n$ ; piecing resampled blocks together builds a bootstrap version of the process  $Z(\cdot)$  on the entire region  $R_n$ . Similar to the MBB, we formulate two block bootstrap variance estimators based on two different sources of blocks for resampling: rectangular subregions  $\mathcal{B}_n(\cdot)$  within  $R_n$  that are overlapping (OL) or non-overlapping (NOL). We describe the OL version in detail; the NOL version is similar.

### 2.2.1 Block bootstrap variance estimator

Define an integer index set  $\mathcal{I}_n = \{\mathbf{i} \in \mathbb{Z}^d : \mathcal{B}_n(\mathbf{i}) \subset R_n\}$  for those integer-translated blocks  $b_n\mathcal{U}$  lying completely inside  $R_n$ ; see Figure 1(b). For each  $\mathbf{k} \in \mathcal{K}_n$ , we resample one block at random from the collection of OL blocks  $\{\mathcal{B}_n(\mathbf{i}) : \mathbf{i} \in \mathcal{I}_n\}$ , independently from other resampled blocks, to define a version of the process  $\{Z(\mathbf{s}) : \mathbf{s} \in R_n(\mathbf{k}) \cap \mathbf{Z}^d\}$  observed on  $R_n(\mathbf{k})$ . To make this precise, let  $\{\mathbf{i}_{\mathbf{k},OL} : \mathbf{k} \in \mathcal{K}_n\}$ , be a collection of iid random variables with common distribution

$$P(\mathbf{i}_{\mathbf{k},OL} = \mathbf{j}) = \frac{1}{|\mathcal{I}_n|}, \quad \mathbf{j} \in \mathcal{I}_n, \quad (3)$$

using  $|A|$  to denote the number of elements in a finite set  $A \subset \mathbb{R}^d$ . For each  $\mathbf{k} \in \mathcal{K}_n$ , the OL block bootstrap version of data  $\mathcal{Z}_n(R_n(\mathbf{k}))$  is defined by  $\mathcal{Z}_{n,OL}^*(R_n(\mathbf{k})) = \mathcal{Z}_n(\mathcal{B}_n(\mathbf{i}_{\mathbf{k},OL}))$ . We concatenate the resampled block observations for each  $\mathbf{k} \in \mathcal{K}_n$  into a single spatial bootstrap sample  $\{\mathcal{Z}_{n,OL}^*(R_n(\mathbf{k})) : \mathbf{k} \in \mathcal{K}_n\}$  defined at sampling sites  $\mathcal{S}_n \cap \{R_n(\mathbf{k}) : \mathbf{k} \in \mathcal{K}_n\}$ , which has  $N_1 \equiv N_{1n} = |\mathcal{K}_n| \cdot b_n^d$  observations; see Figure 1(d). Here  $b_n^d = |b_n\mathcal{U} \cap \mathbf{Z}^d|$  represents the number of sampling sites in a block  $\mathcal{B}_n(\mathbf{i})$ ,  $\mathbf{i} \in \mathcal{I}_n$  or a subregion  $R_n(\mathbf{k})$ ,  $\mathbf{k} \in \mathcal{K}_n$ . Let  $\bar{Z}_{n,OL}^*$  be the average of the  $N_1$  resampled values. We define the OL block bootstrap version of  $\hat{\theta}_n$  as  $\hat{\theta}_{n,OL}^* = H(\bar{Z}_{n,OL}^*)$  and give the corresponding variance estimator of  $\sigma_n^2$  as

$$\hat{\sigma}_{n,OL}^2(b_n) = N_1 \text{Var}_*(\hat{\theta}_{n,OL}^*),$$

where  $\text{Var}_*$  denotes the conditional variance given the data.

Hence, this spatial block resampling scheme is an extension of the OL MBB for time series. In applying the MBB to a size  $n$  time series, the bootstrap time stretch may have marginally shorter length  $n_1 = kb$  because the MBB resamples complete data blocks and ignores some boundary time values in the reconstruction. Analogously with  $\{\mathcal{Z}_{n,OL}^*(R_n(\mathbf{k})) : \mathbf{k} \in \mathcal{K}_n\}$ , we create a bootstrap rendition of those  $R_n$ -observations belonging to some “complete” block  $R_n(\mathbf{k})$ ,  $\mathbf{k} \in \mathcal{K}_n$  in the partition of  $R_n$ ; see Figure 1(d). As in the time series case, some

boundary observations of  $R_n$  may not be used in the bootstrap reconstruction and this occurs commonly with other spatial block methods like subsampling (Sherman, 1996).

Closed-form expressions for  $\hat{\sigma}_{n,OL}^2(b_n)$  are obtainable for some spatial statistics  $\hat{\theta}_n$  (see (7)) while in other cases we may evaluate  $\hat{\sigma}_{n,OL}^2(b_n)$  by Monte-Carlo simulation as follows. Let  $M$  be a large positive integer, denoting the number of bootstrap replicates. For each  $\ell = 1, \dots, M$ , generate a set  $\{\ell \mathbf{i}_{\mathbf{k},OL} : \mathbf{k} \in \mathcal{K}_n\}$ , of iid random variables according to (3) to obtain the  $\ell$ th bootstrap sample *replicate* as  $\{\mathcal{Z}_n(\mathcal{B}_n(\ell \mathbf{i}_{\mathbf{k},OL})) : \mathbf{k} \in \mathcal{K}_n\}$ , whose sample mean evaluated in the function  $H$  yields the  $\ell$ th bootstrap replicate  $\ell \hat{\theta}_{n,OL}^*$  of  $\hat{\theta}_{n,OL}^*$ . The Monte-Carlo approximation to the OL block bootstrap variance estimator is then given by

$$\hat{\sigma}_{n,OL}^2(b_n)^{MC} = \frac{N_1}{M} \sum_{\ell=1}^M \left( \ell \hat{\theta}_{n,OL}^* - E_*^{MC} \hat{\theta}_{n,OL}^* \right)^2, \quad E_*^{MC} \hat{\theta}_{n,OL}^* = \frac{1}{M} \sum_{\ell=1}^M \ell \hat{\theta}_{n,OL}^*.$$

For the NOL block bootstrap method, we consider resampling strictly from the NOL collection of blocks  $\{R_n(\mathbf{k}) : \mathbf{k} \in \mathcal{K}_n\}$ ; see Figure 1(c). Let  $\{\mathbf{i}_{\mathbf{k},NOL} : \mathbf{k} \in \mathcal{K}_n\}$  denote iid random variables with common distribution  $P(\mathbf{i}_{\mathbf{k},NOL} = \mathbf{j}) = 1/|\mathcal{K}_n|$ ,  $\mathbf{j} \in \mathcal{K}_n$ . The NOL bootstrap version of  $\mathcal{Z}_n(R_n(\mathbf{k}))$ ,  $\mathbf{k} \in \mathcal{K}_n$ , is then  $\mathcal{Z}_{n,NOL}^*(R_n(\mathbf{k})) = \mathcal{Z}_n(R_n(\mathbf{i}_{\mathbf{k},NOL}))$ . As before, we combine the resampled NOL block observations into a size- $N_1$  bootstrap sample  $\{\mathcal{Z}_{n,NOL}^*(R_n(\mathbf{k})) : \mathbf{k} \in \mathcal{K}_n\}$  with an average denoted by  $\bar{Z}_{n,NOL}^*$ . The NOL block bootstrap variance estimator of  $\sigma_n^2$  is given by  $\hat{\sigma}_{n,NOL}^2(b_n) = N_1 \text{Var}_*(\hat{\theta}_{n,NOL}^*)$  based on the NOL block bootstrap version  $\hat{\theta}_{n,NOL}^* = H(\bar{Z}_{n,NOL}^*)$  of  $\hat{\theta}_n$ .

## 3 Main results

### 3.1 Assumptions

To describe the results on the spatial block bootstrap, we require some notation and assumptions. For a vector  $\mathbf{x} = (x_1, \dots, x_d)' \in \mathbb{R}^d$ , let  $\|\mathbf{x}\|$ ,  $\|\mathbf{x}\|_1 = \sum_{i=1}^d |x_i|$  and  $\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq d} |x_i|$  denote the Euclidean,  $l^1$  and  $l^\infty$  norms of  $\mathbf{x}$ , respectively. Define the distance between two sets  $E_1, E_2 \subset \mathbb{R}^d$  as:  $\text{dis}(E_1, E_2) = \inf\{\|\mathbf{x} - \mathbf{y}\|_\infty : \mathbf{x} \in E_1, \mathbf{y} \in E_2\}$ . For an uncountable set  $A \subset \mathbb{R}^d$ ,  $\text{vol}(A)$  will refer to the volume (i.e., the  $\mathbb{R}^d$  Lebesgue measure) of  $A$ .

The assumptions below, resembling those in Nordman and Lahiri (2004), include a mixing/moment condition (Assumption 4<sub>r</sub>) stated as a function of a positive argument  $r \in \mathbb{Z}_+ = \{0, 1, 2, \dots\}$ . For  $\nu = (\nu_1, \dots, \nu_p)' \in \mathbb{Z}_+^p$ , let  $D^\nu$  denote the  $\nu$ th order partial differential operator  $\partial^{\nu_1+\dots+\nu_p}/\partial x_1^{\nu_1} \dots \partial x_p^{\nu_p}$  and  $\nabla = (\partial H(\mu)/\partial x_1, \dots, \partial H(\mu)/\partial x_p)'$  be the vector of first order partial derivatives of  $H$  at  $\mu$ . Let  $\mathcal{F}_Z(T)$  denote the  $\sigma$ -field generated

by the variables  $\{Z(\mathbf{s}) : \mathbf{s} \in T\}$ ,  $T \subset \mathbf{Z}^d$  and define the strong mixing coefficient for the random field  $Z(\cdot)$  as  $\alpha(k, l) = \sup\{\tilde{\alpha}(T_1, T_2) : T_i \subset \mathbf{Z}^d, |T_i| \leq l, i = 1, 2; \text{dis}(T_1, T_2) \geq k\}$  where  $\tilde{\alpha}(T_1, T_2) = \sup\{|P(A \cap B) - P(A)P(B)| : A \in \mathcal{F}_Z(T_1), B \in \mathcal{F}_Z(T_2)\}$ .

**Assumptions:**

- (1.) As  $n \rightarrow \infty$ ,  $b_n^{-1} + b_n^{(d+1)/d}/\lambda_n \rightarrow 0$  for  $\lambda_n$  in (1). For any positive sequence  $a_n \rightarrow 0$ , the number of cubes  $a_n(\mathbf{i} + [0, 1)^d)$ ,  $\mathbf{i} \in \mathbb{Z}^d$ , intersecting both  $R_0$  and  $\mathbb{R}^d \setminus R_0$  is  $O(a_n^{-(d-1)})$ .
- (2.)  $H : \mathbb{R}^p \rightarrow \mathbb{R}$  is twice continuously differentiable and, for some  $a \in \mathbb{Z}_+$  and real  $\mathcal{C} \geq 0$ , it holds that  $\max\{|D^\nu H(\mathbf{x})| : \|\nu\|_1 = 2\} \leq \mathcal{C}(1 + \|\mathbf{x}\|^a)$ ,  $\mathbf{x} \in \mathbb{R}^p$ .
- (3.)  $\sigma^2 = \sum_{\mathbf{k} \in \mathbb{Z}^d} \varrho(\mathbf{k}) \in (0, \infty)$ , where  $\varrho(\mathbf{k}) = \text{Cov}(\nabla' Z(\mathbf{t}), \nabla' Z(\mathbf{t} + \mathbf{k}))$ .
- (4<sub>r</sub>.) There exist nonnegative functions  $\alpha_1(\cdot)$ ,  $g(\cdot)$  such that  $\alpha(k, l) \leq \alpha_1(k)g(l)$ . For some  $0 < \delta \leq 1$ ,  $0 < \kappa < (2r - 1 - 1/d)(2r + \delta)/\delta$  and  $\mathcal{C} > 0$ , it holds that  $E\|Z(\mathbf{t})\|^{2r+\delta} < \infty$ ,  $\sum_{m=1}^{\infty} m^{(2r-1)d-1} \alpha_1(m)^{\delta/(2r+\delta)} < \infty$ ,  $g(x) \leq \mathcal{C}x^\kappa$ ,  $x \in [1, \infty)$ .

Growth rates for the blocks  $b_n$  and sampling region  $R_n = \lambda_n R_0$  in Assumption 1 are the spatial analog of scaling used with the MBB for time series  $d = 1$  (Lahiri, 1996). The condition on  $R_0$  is satisfied by most regions of practical interest and holds in the plane  $d = 2$ , for example, if the boundary  $\partial R_0$  of  $R_0$  is delineated by a simple curve of finite length (Sherman, 1996). The  $R_0$ -condition implies that the effect of data points lying near the boundary of  $R_n$  is negligible compared to the totality of data points and that the volume of  $R_n$  determines the number of data points and blocks (see Lemma 1(c) in Appendix). Conditions on the smooth model function  $H$  in Assumption 2 are standard in this context. Assumption 3 implies that a positive limiting variance  $\sigma^2 = \lim_{n \rightarrow \infty} \sigma_n^2$  exists. Assumption 4<sub>r</sub> describes a bound on the strong mixing coefficient satisfying certain growth conditions. Bounds of this type are known to apply for many weakly dependent random fields and time series; see Doukhan (1994) and Guyon (1995). This assumption also allows moment bounds for sample means of the process  $Z(\cdot)$  as well as a central limit theorem (Lahiri, 2003b).

### 3.2 Bias and variance expansions

To find optimal block sizes, we first provide the asymptotic bias and variance of the spatial block bootstrap estimators  $\hat{\sigma}_{n,OL}^2(b_n)$  and  $\hat{\sigma}_{n,NOL}^2(b_n)$ .

**Theorem 1** (i) Suppose that Assumptions 1 - 3 and 4<sub>2+2a</sub> hold with “a” as specified under



Assumption 2 and  $B_0 \neq 0$  in (4). Then, the bias of  $\hat{\sigma}_n^2(b_n)$  is

$$\mathbb{E}[\hat{\sigma}_n^2(b_n) - \sigma_n^2] = -\frac{B_0}{b_n}(1 + o(1)), \quad B_0 = \sum_{\mathbf{k} \in \mathbb{Z}^d} \|\mathbf{k}\|_1 \varrho(\mathbf{k}) \in \mathbb{R}, \quad (4)$$

where  $\hat{\sigma}_n^2(b_n)$  represents either  $\hat{\sigma}_{n,OL}^2(b_n)$  or  $\hat{\sigma}_{n,NOL}^2(b_n)$ .

(ii) If Assumptions 1-3 and  $4_{5+4a}$  hold with “a” as specified under Assumption 2, then

$$\text{Var}[\hat{\sigma}_{n,OL}^2(b_n)] = \left(\frac{2}{3}\right)^d \text{Var}[\hat{\sigma}_{n,NOL}^2(b_n)](1 + o(1)), \quad \text{Var}[\hat{\sigma}_{n,NOL}^2(b_n)] = \frac{2\sigma^4 b_n^d}{\text{vol}(R_n)}(1 + o(1)).$$

Because more OL blocks are generally available than NOL ones, the variance of OL block estimator turns out to be  $(2/3)^d$  times smaller than the variance of the NOL block estimator. For  $d = 1$ , a size  $n$  time series sample is obtained from (1) by setting  $R_0 = (-1/2, 1/2]$ ,  $\lambda_n = n$  on the untranslated lattice  $\mathbf{Z} = \mathbb{Z}$  and applying this in Theorem 1 gives the well-known bias/variance of the MBB variance estimator with time series (Künsch, 1989, Hall *et al.*, 1995, Lahiri, 1996).

We also note some important differences between the spatial bootstrap and spatial subsampling as an alternative for variance estimation with subregions of dependent data (Sherman and Carlstein, 1994, Sherman, 1996). While the block bootstrap aims to recreate a “size  $R_n$ ” sampling region, subsampling considers scaled-down copies  $b_n R_0$  of  $R_n = \lambda_n R_0$  that can be treated as repeated, smaller versions of  $R_n$  on which to evaluate a spatial statistic  $\hat{\theta}_n$  (Politis *et al.*, 1999, Section 3.8); the subsampling variance estimator of  $\text{Var}(\hat{\theta}_n)$  is the sample variance of the subsample evaluations of  $\hat{\theta}_n$ . Philosophical differences in the bootstrap/subsampling mechanics translate into large differences in their bias and variance expansions. For example, if  $\bar{Z}_{\mathbf{0},n}$  denotes the sample mean over the  $b_n^d$  observations in the block  $b_n \mathcal{U}$  for the bootstrap or over the  $|b_n R_0 \cap \mathbf{Z}^d|$  observations in  $b_n R_0$  for subsampling, the expected values  $\mathbb{E}\hat{\sigma}_n^2$  of bootstrap/subsampling variance estimators of  $\sigma_n^2 = N\text{Var}(\hat{\theta}_n)$  are

$$\text{bootstrap: } b_n^d \text{Var}(\nabla' \bar{Z}_{\mathbf{0},n}) + o(N^{-1/2}) \quad \text{subsampling: } |b_n R_0 \cap \mathbf{Z}^d| \text{Var}\{H(\bar{Z}_{\mathbf{0},n})\}(1 + o(1)).$$

That is, for *any* dimension  $d$  of sampling, the bootstrap bias (i.e.,  $\mathbb{E}\hat{\sigma}_n^2 - \sigma_n^2$ ) is determined by the first (linear) term  $\nabla' \bar{Z}_{\mathbf{0}}$  in a Taylor’s expansion of  $H(\bar{Z}_{\mathbf{0},n})$  around  $\mu$ . This is not generally true for subsampling, which can require up to four terms in the Taylor’s expansion of  $H(\bar{Z}_{\mathbf{0},n})$  to pinpoint the main  $O(b_n^{-1})$  bias component, especially in lower dimensions  $d = 1, 2$ ; see Nordman and Lahiri (2004). Under appropriate conditions, subsampling variance estimators have the same variance and bias orders (i.e.,  $b_n^d/\lambda_n^d, b_n^{-1}$ ) as the bootstrap, but the proportionality constants for these orders depend greatly on the geometry of  $R_n$  and are not



as simple as the bootstrap expressions in Theorem 1. This can make plug-in estimation of optimal block sizes (see Section 5) generally more difficult for subsampling compared to the bootstrap.

### 3.3 Theoretical optimal block size

The bias and variance expansions from Theorem 1 yield an asymptotic MSE (e.g.,  $E[\hat{\sigma}_n^2(b_n) - \sigma_n^2]^2$ ), which the theoretically best block size  $b_n$  minimizes to optimize the over-all performance of a block bootstrap variance estimator. Explicit expressions for optimal OL and NOL block sizes for block bootstrap variance estimation are given in the following. We find that, in large samples, OL blocks should be larger than NOL ones by a factor owing to differing variances in Theorem 1.

**Theorem 2** *Under the assumptions of Theorem 1(ii), the optimal block sizes  $b_{n,OL}^{\text{opt}}$  and  $b_{n,NOL}^{\text{opt}}$  for  $\hat{\sigma}_{n,OL}^2(\cdot)$  and  $\hat{\sigma}_{n,NOL}^2(\cdot)$  are given by*

$$b_{n,OL}^{\text{opt}} = b_{n,NOL}^{\text{opt}} \left( \frac{3}{2} \right)^{d/(d+2)} (1 + o(1)), \quad b_{n,NOL}^{\text{opt}} = \left( \frac{B_0^2 \cdot \text{vol}(R_n)}{d\sigma^4} \right)^{1/(d+2)} (1 + o(1)). \quad (5)$$

**Remark:** The volume  $\text{vol}(R_n)$  may be replaced by the sample size  $N$  in Theorems 1-2.

In terms of the spatial sample size, the optimal block size for bootstrap variance estimation is  $O(N^{1/(d+2)})$ . For the time series case  $d = 1$ , we set the time series length  $n$  for  $\text{vol}(R_n)$  in Theorem 2 to yield the optimal block expression obtained by other authors with  $\sigma^2 = \sum_{\mathbf{k}=-\infty}^{\infty} \varrho(\mathbf{k})$ ,  $B_0 = \sum_{\mathbf{k}=-\infty}^{\infty} |\mathbf{k}| \varrho(\mathbf{k})$  (Künsch, 1989, Hall *et al.*, 1995); see Section 3.2 for defining  $R_n$  with time series. Hence, there is continuity in the optimal blocks between spatial and times series bootstrap settings. In the spatial case, the key difference is that optimal blocks depend on the dimension  $d$  of sampling as well as the volume of the sampling region  $R_n$  (or relatedly the spatial sample size) as determined by the geometry of  $R_n$ .

In subsequent sections, we turn to examining practical methods for estimating the optimal block size for using the spatial bootstrap.

## 4 Empirical choice of the optimal block size

We describe two general approaches for estimating the optimal block size for the bootstrap estimators. These are spatial versions of block selection procedures for time series and involve either cross-validation (Hall *et al.*, 1995) or “plug-in” estimators. Let  $\hat{\sigma}_n(\cdot)$  denote either the OL or NOL block-based variance estimator in the following.

Hall *et al.* (1995) proposed a method for choosing a block length with the time series MBB, in which a type of subsampling is used to estimate the MSE incurred by the bootstrap at various block sizes. To describe a spatial version of their procedure, let  $m \equiv m_n$  be a positive integer sequence satisfying  $m^{-1} + m/\lambda_n \rightarrow 0$  and define a set  $\mathcal{M}_m = \{\mathbf{i} \in \mathbb{Z}^d : \mathbf{i} + m\mathcal{U} \subset R_n\}$  of all OL cubes  $m\mathcal{U}$  (of side length  $m$ ) within  $R_n = \lambda_n R_0$ . Consider the rectangular subsamples of the data:  $\mathcal{Z}_n(\mathbf{i} + m\mathcal{U})$ ,  $\mathbf{i} \in \mathcal{M}_m$ . (The subsampling regions here correspond to cubes lying inside  $R_n$ , not scaled-down copies of  $R_n$  more typically associated with subsampling (Sherman, 1996).) Let  $\hat{\sigma}_{\mathbf{i},m}^2(b)$ ,  $\mathbf{i} \in \mathcal{M}_m$ , denote the block bootstrap variance estimator applied to each subsample  $\mathcal{Z}_n(\mathbf{i} + m\mathcal{U})$  by resampling blocks of size  $b$ . A subsampling estimator of  $\text{MSE}(\hat{\sigma}_{\mathbf{i},m}^2(b))$ , the mean squared error of the bootstrap variance estimator on a rectangular subsample, is then given as

$$\widehat{\text{MSE}}_m(b) = \frac{1}{|\mathcal{M}_m|} \sum_{\mathbf{i} \in \mathcal{M}_m} \left[ \hat{\sigma}_{\mathbf{i},m}^2(b) - \hat{\sigma}_n^2(b_n^*) \right]^2 \quad (6)$$

where  $b_n^*$  is a pilot block size. For some  $\epsilon > 0$ , let  $\mathcal{J}_{m,\epsilon} = \{b \in \mathbb{Z}_+ : m^{d/(d+2)-\epsilon} \leq b \leq m^{d/(d+2)+\epsilon}\}$  and define the minimizer of (6) as  $\hat{b}_m^0 = \arg \min \{\widehat{\text{MSE}}_m(b) : b \in \mathcal{J}_{m,\epsilon}\}$ , where  $\hat{b}_m^0$  estimates the optimal block size for a sampling region of size  $\text{vol}(m\mathcal{U}) = m^d$ . By Theorem 3, we may re-scale  $\hat{b}_m^0$  to obtain a block size estimator for the original sampling region  $R_n$  by

$$\hat{b}_n^{\text{HHJ}} = \hat{b}_m^0 \cdot [\text{vol}(R_n)/m^d]^{1/(d+2)}.$$

We refer to this as HHJ method.

A second, computationally simpler estimator of the best block size is through a non-parametric plug-in (NPI) method. This involves directly substituting estimates of population quantities into the theoretical block expressions from (5). For the time series MBB, plug-in choices of block size have been suggested in various forms by Bühlmann and Künsch (1999), Politis and White (2003), and Lahiri *et al.* (2006). Our NPI approach for the spatial block size involves two integer sequences  $b_{n,1}^{\text{NPI}}$  and  $b_{n,2}^{\text{NPI}}$  of block sizes satisfying  $1/b_{n,i}^{\text{NPI}} + b_{n,i}^{\text{NPI}}/\lambda_n^{d/(d+i)} \rightarrow 0$  as  $n \rightarrow \infty$ , for  $i = 1, 2$ . Following Lahiri *et al.* (2006), we use the difference of two bootstrap variance estimators to estimate the bias component  $B_0$  from (5) by  $\widehat{B}_0 = 2b_{n,2}^{\text{NPI}} \cdot [\hat{\sigma}_n^2(b_{n,2}^{\text{NPI}}) - \hat{\sigma}_n^2(2b_{n,2}^{\text{NPI}})]$ . The expectation result in Theorem 1 suggests  $\widehat{B}_0$  as an appropriate estimator of  $B_0$ . Using the second block size  $b_{n,1}^{\text{NPI}}$ , we estimate the spread component  $\sigma^2$  in (5) with  $\hat{\sigma}^2 = \hat{\sigma}_n^2(b_{n,1}^{\text{NPI}})$ . Substituting  $\widehat{B}_0$  and  $\hat{\sigma}^2$  into the theoretical expression from Theorem 2 gives a NPI estimator  $\hat{b}_n^{\text{NPI}}$  of the optimal block size for  $\hat{\sigma}_n^2(\cdot)$ .

Both HHJ and NPI block selection methods are consistent under mild conditions. In previous work, Hall *et al.* (1995) did not establish the consistency of their block method with

the time series MBB and the method's formal consistency has remained an open problem. We provide the first consistency result for the HHJ method (to our knowledge), which is valid for both spatial and time series data. In the following, let  $b_n^{\text{opt}}$  denote the theoretical optimal block size for  $\hat{\sigma}_n^2(\cdot)$  and suppose  $\hat{\sigma}_n^2(\cdot)$  is used to compute  $\hat{b}_n^{\text{HHJ}}$  or  $\hat{b}_n^{\text{NPI}}$ .

**Theorem 3** *Assume  $m^{2d+\epsilon(d+2)}/\text{vol}(R_n) = o(1)$  for  $\epsilon$  defining  $\mathcal{J}_{m,\epsilon}$ ,  $b_n^* = C^*\text{vol}(R_n)^{1/(d+2)}$  for some  $C^* > 0$ . In addition to Assumptions 1-3 and  $B_0 \neq 0$ , suppose Assumption 4<sub>r</sub> holds using  $r = 5 + 4a$  for  $\hat{b}_n^{\text{NPI}}$  or  $r = 15 + 12a$  for  $\hat{b}_n^{\text{HHJ}}$  (with  $a$  as specified under Assumption 2). Then as  $n \rightarrow \infty$ ,  $\hat{b}_n^{\text{NPI}}/b_n^{\text{opt}} \xrightarrow{p} 1$  and  $\hat{b}_n^{\text{HHJ}}/b_n^{\text{opt}} \xrightarrow{p} 1$ .*

The HHJ block estimator  $\hat{b}_n^{\text{HHJ}}$  depends on the subsampling tuning parameter  $m$  and a block size  $b_n^* = C^*\text{vol}(R_n)^{1/(d+2)}$ ,  $C^* > 0$  (e.g.,  $C^* = 1, 2$ ), of the optimal order from Theorem 2. As in the time series case, the optimal value of  $m$  is unknown in the spatial setting. However, to reduce the effect of the tuning parameter  $b_n^*$ , we may follow the iteration proposal of Hall *et al.* (1995). Upon obtaining an estimate  $\hat{b}_n^{\text{HHJ}}$ , this value may be set as the pilot block  $b_n^*$  in a second round of the algorithm and this iterative process may be repeated until convergence of the block estimate. This is the approach we apply in Section 5.

For the NPI estimator  $\hat{b}_n^{\text{NPI}}$  requiring smoothing parameters  $b_{n,i}^{\text{NPI}}$ ,  $i = 1, 2$ , it can be established that the optimal order of the block size  $b_{n,2}^{\text{NPI}}$  for estimating the bias  $B_0$  is  $\text{vol}(R_n)^{1/(d+4)}$  while the optimal order of the block size  $b_{n,1}^{\text{NPI}}$  is  $\text{vol}(R_n)^{1/(d+2)}$  from Theorem 2. Hence, plausible tuning parameters have the form  $b_{n,i}^{\text{NPI}} = C_i \text{vol}(R_n)^{1/(d+2i)}$  for  $C_i > 0$ ,  $i = 1, 2$ .

## 5 Simulation Study

In this section, we summarize numerical studies of the performance of spatial block bootstrap variance estimators as well as empirical methods for choosing the block size.

### 5.1 Block Bootstrap MSE and Optimal Blocks

We first conducted a simulation study to compare OL and NOL block bootstrap variance estimators of  $\sigma_n^2 = N\text{Var}(\hat{\theta}_n)$  as well as a spatial subsampling estimator, where  $\hat{\theta}_n = \bar{Z}_N$  is the sample mean over a circular sampling region  $R_n$ . Two regions of different sizes were considered  $R_n := \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 9\}$  and  $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 20\}$ . We used the circulant embedding method of Chan and Wood (1997) to generate real-valued mean-zero Gaussian random fields on  $\mathbf{Z}^2 = \mathbb{Z}^2$  (i.e.,  $\mathbf{t} = \mathbf{0}$ ) with an Exponential covariance structure:

$$\text{Model}(\beta_1, \beta_2) : \varrho(\mathbf{k}) = \exp \left[ -\beta_1 |k_1| - \beta_2 |k_2| \right], \quad \mathbf{k} = (k_1, k_2)' \in \mathbb{Z}^2, \quad \beta_1, \beta_2 > 0.$$

We considered the values  $(\beta_1, \beta_2) = (1, 1)$  and  $(0.5, 0.3)$  to obtain covariograms exhibiting various rates of decay. Because the smooth model function is linear here (i.e.,  $H(x) = x$ ), the bootstrap estimator  $\hat{\sigma}_{n,OL}^2(b_n)$  of  $\sigma_n^2$  has a closed form expression as a scaled sample variance of block means

$$\hat{\sigma}_{n,OL}^2(b_n) = |\mathcal{I}_n|^{-1} \sum_{\mathbf{i} \in \mathcal{I}_n} b_n^d \left( \bar{Z}_{\mathbf{i},n} - \bar{Z}_n \right)^2, \quad \bar{Z}_n = |\mathcal{I}_n|^{-1} \sum_{\mathbf{i} \in \mathcal{I}_n} \bar{Z}_{\mathbf{i},n}, \quad (7)$$

where  $\bar{Z}_{\mathbf{i},n}$  denotes the sample mean of the  $b_n^d$  ( $d = 2$ ) observations in an OL block  $\mathcal{B}_n(\mathbf{i})$ ,  $\mathbf{i} \in \mathcal{I}_n$ ;  $\hat{\sigma}_{n,NOL}^2(b_n)$  is given by replacing  $\mathcal{I}_n$ ,  $\bar{Z}_{\mathbf{i},n}$  with NOL analogs  $\mathcal{K}_n$ ,  $\bar{Z}_{b_n \mathbf{i},n}$ .

For comparison, we also evaluated the OL subsampling variance estimator of the sample mean  $\hat{\theta}_n = \bar{Z}_n$  with subsamples as translates of  $b_n R_0$ , based on the template  $R_0 = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1/2\}$  for  $R_n$ . Here the OL subsampling estimator of  $\sigma_n^2$  has a form similar to  $\hat{\sigma}_{n,OL}^2$  but employs OL subcircles rather than cubes (i.e., redefine  $\mathcal{I}_n$ ,  $b_n^d$  in (7) by using  $b_n R_0$  instead of  $b_n \mathcal{U}$ ). That is, in the case of the sample mean, the only difference between the OL block bootstrap and subsampling variance estimators is the subregion shape used and the bootstrap employs rectangular blocks.

Figure 2 gives the normalized MSEs for each variance estimator over various block sizes with optimal block sizes also denoted. We observe the following from the results:

1. The OL bootstrap estimator generally performed as good as (and often better than) than the NOL version at *any* fixed block size considered.
2. At their optimal blocks, the OL block bootstrap performed slightly better than OL subsampling in most cases. Table 1 provides a decomposition of the bias and variance of both estimators at optimal block scaling. In the one instance where subsampling outperformed the bootstrap (i.e., Model(0.5,0.3) on the larger  $R_n$ ), more optimal OL circular subsamples were available than OL optimal blocks (i.e., 665 subcircles compared to 632 blocks).
3. The MSEs for the OL bootstrap exhibited more curvature than subsampling as a function of block size, which may suggest a possible advantage for the bootstrap in estimating/locating optimal block sizes. This topic is beyond the scope of the current paper but may be important for future investigation.

## 5.2 Block Selection Methods

Using the same regions  $R_n$  and covariance models, we next studied empirical block selection for the OL block bootstrap by investigating possible values for constants  $(C_1, C_2)$

and  $(C^*, m)$ , used to define the tuning parameters  $\hat{b}_{n,i}^{\text{NPI}} = C_i \text{vol}(R_n)^{1/(d+2i)}$ ,  $i = 1, 2$  and  $b_n^* = C^* \text{vol}(R_n)^{1/(d+2)}$  in the NPI and HHJ methods, respectively. With time series, numerical studies for the MBB suggest that constants near one perform adequately for plug-in block choices (Lahiri *et al.*, 2006). Hence, we considered combinations of scalars  $C_1, C_2, C^* \in [1/2, 2]$  in increments of  $1/4$  and values  $m = \{C \text{vol}(R_n)^{1/6} : C = 2, 3, 4\}$  to produce an estimator  $\hat{b}_{n,OL}$  of the optimal OL block size  $b_{n,OL}^{\text{opt}}$  with the NPI or HHJ approach. (The  $m$ -values in the HHJ method were chosen under Theorem 3 with  $\epsilon = 1/2$  and  $\mathcal{J}_{m,\epsilon} = [1, m] \cap \mathbb{Z}$ ; the resulting values for  $m$  were typically around  $m = 5, 10, 15$ .) To initially assess a block estimator, an MSE-criterion

$$\mathbb{E}\{\hat{\sigma}_{n,OL}^2(\hat{b}_{n,OL}) - \hat{\sigma}_{n,OL}^2(b_{n,OL}^{\text{opt}})\}^2 / \sigma_n^4 \quad (8)$$

was evaluated based on 10,000 simulations for each sampling region  $R_n$ , dependence structure, and  $(C_1, C_2)$  or  $(C^*, m)$  combination. The criterion assesses the discrepancy between bootstrap variance estimators based on a block estimate and the optimal OL block choice from Figure 2.

The simulation results (not included here for brevity) indicated tuning parameters values of  $C_1, C_2 \leq 1$  were generally best for the NPI method, especially with  $C_2 = 1/2$ . For the HHJ method, a constant  $C^* \in [1, 2]$  performed adequately as observed in Politis and Sherman (2001) while values around  $m = 10, 15$  performed best on the larger sampling region (especially under strongest spatial dependence with Model(0.5,0.3) where larger subsample blocks may be expected). Figure 3 illustrates dot plots of the MSE-criterion (8) to give a rough comparison of NPI and HHJ methods over the mentioned tuning parameter ranges where the methods often performed well (i.e.,  $C_1, C_2 \in \{0.5, .75, 1\}$ ,  $C^* \in \{1, 1.5, 2\}$ ). Figure 3 indicates that both NPI and HHJ methods often performed comparably in terms of (8) with no one method emerging as clearly superior. The NPI method often performed well and produced the lowest MSE-realizations but the HHJ exhibited more stability (less variability) across a larger number of tuning parameters. In this sense, the HHJ method appeared more robust to the tuning parameter selection, which may be due to the HHJ iteration steps to minimize the effect of  $C^*$  in  $b_n^*$  (described at the end of Section 4). At the same time, the HHJ method is computationally more intensive than the NPI which can be prohibitive in some applications (see the data example of Section 6.2).

Table 2 displays the frequency distribution of OL block estimates based on both selection methods. Both NPI and HHJ approaches performed fairly well in identifying the optimal OL block size and both methods appear comparable. On the larger sampling regions, the

NPI and HHJ selections seemed to exhibit more variability but empirical block estimates that deviate slightly from the optimal block appear less serious for these regions (since the bootstrap MSEs in Figure 2 are similar around the optimal block size).

## 6 Data examples

### 6.1 Data Example 1: Longleaf Pines

We illustrate the spatial bootstrap method under the smooth function model with longleaf pine data from Cressie (1993). Figure 8.4 of Cressie (1993) provides counts for the number of pines in a  $32 \times 32$  grid of quadrats observed in a forest region of the Wade Tract, Georgia, where each quadrat corresponds to a  $(0, 6.25] \times [0, 6.25)$  m<sup>2</sup> section partitioning a  $(0, 200] \times [0, 200)$  m<sup>2</sup> study region. Analysis by Sherman (1996) suggests that the spatial counts are stationary. We wish to test if the locations of pines exhibit complete spatial randomness or the tendency to cluster through the clustering index  $ICS \equiv s_n^2/\bar{x}_n - 1$  of David and Moore (1954), involving the sample mean  $\bar{x}_n$  and variance  $s_n^2$  of the  $N = 1024$  quadrat counts (defining  $s_n^2$  here with a divisor  $N$ ). A positive ICS-value implies the trees tend to cluster while we should expect the statistic to be zero when the tree locations are completely spatially random; see Cressie (1993) for details.

To judge if the observed  $ICS = 1.164$  for the longleaf pine counts is significantly greater than zero, a test statistic requires a variance estimate of  $\sigma_n^2 = N\text{Var}(ICS)$  that accounts for possible spatial dependence between the counts. The block bootstrap is applicable under the smooth function model. To see this, let  $H : \mathbb{R}^2 \rightarrow \mathbb{R}$  for  $H(x, y) = x^{-1}y - x - 1$  and define  $\bar{Z}_N$  as the bivariate sample mean of  $Z(\mathbf{s}) = (X(\mathbf{s}), X^2(\mathbf{s}))'$  of the tree counts  $X(\mathbf{s})$  over quadrat sites  $\mathbf{s} \in R_n \cap \mathbb{Z}^2$ ,  $R_n = (0, 32]^2$ . Then, the ICS index equals  $\hat{\theta}_n = H(\bar{Z}_N)$ .

Due to the nonlinearity of  $\hat{\theta}_n$ , we evaluated the OL block estimator  $\hat{\sigma}_{n,OL}^2$  of  $\sigma_n^2 = N\text{Var}(\hat{\theta}_n)$  by  $M = 3000$  Monte-Carlo simulations. Both NPI and HHJ methods tended to agree in their block selections. The NPI method selected an OL block  $\hat{b}_{n,OL}^{\text{NPI}} = 3$  using  $C_2 = 0.5$  and  $C_1 \in \{0.5, 1, 2\}$  in the two pilot blocks  $\hat{b}_{n,i}^{\text{NPI}} = C_i N^{1/(d+2i)}$ ,  $i = 1, 2$ ; changing  $C_2 = 2$  with the same  $C_1$  range produced  $\hat{b}_{n,OL}^{\text{NPI}} = 3$  or 5. Using  $b_n^* = b_{n,1}^{\text{NPI}}$  (i.e.,  $C^* = C_1 \in \{0.5, 1, 2\}$ ) in the HHJ method led to block estimates  $\hat{b}_{n,OL}^{\text{HHJ}} = 3$  for  $m \in \{5, 20\}$ ,  $\hat{b}_{n,OL}^{\text{HHJ}} = 5$  for  $m = 10$  and  $\hat{b}_{n,OL}^{\text{HHJ}} = 6$  for  $m = 15$ . That is, both NPI and HHJ appeared to be fairly stable over one tuning parameter  $C_1 = C^*$  in this example and changed slightly over the second tuning parameter ( $C_2$  for NPI or  $m$  for HHJ); note that we may anticipate HHJ to

be robust to  $C^*$  due to iteration steps in the method (see Section 4). The resulting estimates  $\hat{\sigma}_{n,OL}^2(3) = 141.58$ ,  $\hat{\sigma}_{n,OL}^2(5) = 126.60$  and  $\hat{\sigma}_{n,OL}^2(6) = 116.53$  produced estimated standard errors  $(\hat{\sigma}_{n,OL}^2/N)^{1/2}$  of ICS as 0.372, 0.352 and 0.337 that indicate significant evidence of clustering based on a normal approximation for the ICS index. Other findings in Chapter 8 of Cressie (1993) involving Poisson goodness-of-fit tests or distance methods support this conclusion as well as the subsampling analysis by Sherman (1996).

## 6.2 Data Example 2: U.S. Cancer Mortality Map

Sherman and Carlstein (1994) provide a binary mortality map of the United States that indicates “high” or “low” mortality rates from liver and gallbladder cancer in white males during 1950-1959. The map consists of a spatial region  $R_n$  containing 2298 sites on the integer  $\mathbb{Z}^2$  grid, of which 2012 have four nearest-neighbors available; see Sherman and Carlstein (1994) for details. For a given site  $\mathbf{s} \in \mathbb{Z}^2$ , we code data  $Z(\mathbf{s}) = 0$  or 1 to indicate a low or high mortality rate. (Note Sherman and Carlstein (1994) used coding  $-1/1$  rather than  $0/1$ .)

To investigate whether incidences of high cancer mortality exhibit clustering, Sherman and Carlstein (1994) proposed fitting an autologistic model fit to the data as follows. Suppose the data were generated by an autologistic model [Besag (1974)]:

$$P(Z(\mathbf{s}) = z \mid Z(\mathbf{i}), \mathbf{i} \neq \mathbf{s}) = \frac{\exp [z(\alpha + \beta N(\mathbf{s}))]}{1 + \exp [\alpha + \beta N(\mathbf{s})]}, \quad z = 0, 1, \quad (9)$$

where  $N(\mathbf{s}) = \sum_{\|\mathbf{h}-\mathbf{s}\|=1} Z(\mathbf{h})$  denotes the sum of indicators over the four nearest-neighbors of site  $\mathbf{s}$ . Positive values of the parameter  $\beta$  suggest a tendency of clustering, while  $\beta = 0$  implies no clustering among sites. Maximum pseudolikelihood estimation (MPLE) with the  $N = 2012$  pairs  $Z(\mathbf{s}), N(\mathbf{s})$  yielded  $\hat{\beta} = 0.419$  and  $\hat{\alpha} = -1.485$ . To test if the estimate  $\hat{\beta}$  is significantly greater than zero, we require an estimate of  $N\text{Var}(\hat{\beta})$  that accounts for spatial dependence because standard errors for MPLE statistics  $\hat{\beta}$  generally have no closed-forms.

We evaluated the OL block bootstrap estimator of  $N\text{Var}(\hat{\beta})$  based on  $M = 800$  Monte-Carlo simulations. For this variance estimation problem, a NPI estimate of OL block size is relatively fast and easy to compute while the HHJ method is computationally prohibitive (requiring many iterations of MPLE to be applied in the subsampling steps). Following tuning parameters from Section 5.2, we set  $C_1 = C_2 = 0.5$  for  $\hat{b}_{n,i}^{\text{NPI}} = C_i N^{1/(d+2i)}$ ,  $i = 1, 2$ , in the NPI method which selected an OL block  $\hat{b}_{n,OL}^{\text{NPI}} = 4$  with associated variance estimate  $\hat{\sigma}_{n,OL}^2(4) = 12.84$ . The corresponding estimated standard error  $(\hat{\sigma}_{n,OL}^2(4)/N)^{1/2}$  of  $\hat{\beta}$  is 0.0799, from which we conclude significant evidence of clustering. This agrees with other analyses



performed by Sherman and Carlstein (1994) accounting for their parametrization. Similar NPI blocks  $\hat{b}_{n,OL}^{NPI} \in \{2, 4, 5\}$  with standard error estimates around 0.08 also followed from other tuning combinations  $C_1, C_2 \in \{0.5, 0.75, 1\}$  in this example.

Because the MPLE statistic  $\hat{\beta}$  does not readily fit into the smooth model framework (Section 2.1), we further examined the bootstrap variance estimator  $\hat{\sigma}_{n,OL}^2$  and block selection in a small, but computationally involved, numerical study. We simulated data from an autologistic model (9) with  $\alpha = -1.49$ ,  $\beta = 0.42$  on a region of the same shape as the U.S. mortality map. (For a single simulated data set, we generated independent Bernoulli  $p = 0.1$  variables on the grid and applied the Gibbs Sampler with a burn-in of 10,000 iterations.) The mean  $E\hat{\beta}$  and variance  $N\text{Var}(\hat{\beta})$  of the MPLE statistic were found to be 0.419 and 11.835 based on 10,000 simulated data sets. Table 3 displays the MSEs associated with the OL block estimator  $\hat{\sigma}_{n,OL}^2$  (based on  $M = 800$  Monte-Carlo simulations) for various blocks  $b_n$ . The table also shows the distribution of OL block size estimates using the NPI method with  $C_1 = C_2 = 0.5$ . The median and mode of the block estimates equal the optimal block length  $b_{n,OL}^{\text{opt}}=3$  and simulations with other choices of  $C_1, C_2 \in [1/2, 2]$  led to similar results.

## 7 Appendix: Proofs of Main Results

To save space, we consider only the OL block bootstrap estimator  $\hat{\sigma}_{n,OL}^2(b_n)$  in detail. Proofs for the NOL block version follow in an analogous manner. For  $\mathbf{i} \in \mathcal{I}_n$  and  $\mathbf{k} \in \mathcal{K}_n$ , let  $U_{\mathbf{i}}$  and  $U_{\mathbf{k}}^*$  respectively denote the sample averages of the  $b_n^d$  observations in  $\mathcal{Z}_n(\mathcal{B}_n(\mathbf{i}))$  and  $\mathcal{Z}_{n,OL}^*(R_n(\mathbf{k}))$ . Define  $\hat{\mu}_n = |\mathcal{I}_n|^{-1} \sum_{\mathbf{i} \in \mathcal{I}_n} U_{\mathbf{i}}$ , a centered version  $\hat{\mu}_{n,cen} = \hat{\mu}_n - \mu$ ,  $\hat{\nabla} = (\partial H(\hat{\mu}_n)/\partial x_1, \dots, \partial H(\hat{\mu}_n)/\partial x_p)'$ ,  $W_{\mathbf{i},cen} = \hat{\nabla}'(U_{\mathbf{i}} - \mu)$  and  $W_{\mathbf{k}}^* = \hat{\nabla}'U_{\mathbf{k}}^*$ . For  $\nu = (\nu_1, \dots, \nu_p)' \in (\mathbb{Z}_+)^p$ ,  $\mathbf{x} \in \mathbb{R}^p$ , write  $\mathbf{x}^\nu = \prod_{i=1}^p x_i^{\nu_i}$ ,  $\nu! = \prod_{i=1}^p (\nu_i!)$ ,  $c_\nu = D^\nu H(\mu)/\nu!$  and  $\hat{c}_\nu = D^\nu H(\hat{\mu}_n)/\nu!$ . We assume that the  $\mathbb{R}^d$  zero vector  $\mathbf{0} \in \mathcal{K}_n$ . Let  $\mathcal{C}$  to denote a generic positive constant that does not depend on  $n$ , block sizes, or any  $\mathbb{Z}^d$ ,  $\mathbf{Z}^d$  points. Appearances of “ $a, r$ ” refer to the  $a, r$ -values from Assumptions 2 and 4 $_r$  for the theorem under consideration. Limits in order symbols are taken letting  $n$  tend to infinity.

We require some results in Lemma 1, where parts(a)-(b) follow from Doukhan (1994, p. 9, 26) and Jensen’s inequality and part(c) holds by the  $R_0$ -boundary condition in Assumption 1.

**Lemma 1** (a) Suppose  $T_1, T_2 \subset \mathbf{Z}^d$  are bounded with  $\text{dis}(T_1, T_2) > 0$ . Let  $p, q > 0$  where  $1/p + 1/q < 1$ . If the random variable  $X_i$  is measurable with respect to  $\mathcal{F}_Z(T_i)$ ,  $i = 1, 2$ , then  $|\text{Cov}(X_1, X_2)| \leq 8(E|X_1|^p)^{1/p}(E|X_2|^q)^{1/q}\alpha(\text{dis}(T_1, T_2); \max_{i=1,2} |T_i|)^{1-1/p-1/q}$ .

(b) Under Assumption 4 $_r$ ,  $r \in \mathbb{Z}_+$ , it holds that for  $1 \leq m \leq 2r$  and any  $T \subset \mathbf{Z}^d$ ,

$E\|\sum_{\mathbf{s} \in T} (Z(\mathbf{s}) - \mu)\|^m \leq \mathcal{C}|T|^{m/2}$  and  $E\|\hat{\mu}_n - \mu\|^m \leq \mathcal{C}N^{-m/2}$ .

(c) Under Assumption 1,  $N/\text{vol}(R_n) \rightarrow 1$ ,  $|\mathcal{K}_n|/(b_n^{-d}\text{vol}(R_n)) \rightarrow 1$ ,  $|\mathcal{I}_n|/\text{vol}(R_n) \rightarrow 1$ .

To prove Theorem 1, we write  $\hat{\sigma}_{n,OL}^2(b_n) = \sum_{i=1}^2 \text{Var}_*(E_{in}^*) + 2\text{Cov}_*(E_{1n}^*, E_{2n}^*)$  where  $E_{1n}^* = \sum_{\|\nu\|_1=1} \hat{c}_\nu(\bar{Z}_{n,OL}^* - \hat{\mu}_n)^\nu$ ,  $E_{2n}^* = 2 \sum_{\|\nu\|_1=2} (\nu!)^{-1} (\bar{Z}_{n,OL}^* - \hat{\mu}_n)^\nu \int_0^1 (1-\omega) D^\nu H(\hat{\mu}_n + \omega(\bar{Z}_{n,OL}^* - \hat{\mu}_n)) d\omega$ , using a Taylor expansion of  $\hat{\theta}_{n,OL}^* = H(\bar{Z}_{n,OL}^*)$  around  $\hat{\mu}_n$ .

**Proof of Theorem 1(i).** By Assumption 1 and Lemma 1(c), it suffices to show that:

$$E(N_1 \text{Var}_*(E_{1n}^*)) - \sigma_n^2 = -\frac{B_0}{b_n}(1 + o(1)), \quad E(N_1 \text{Var}_*(E_{2n}^*)) = O(N^{-1}), \quad (10)$$

from which  $N_1 E|\text{Cov}_*(E_{1n}^*, E_{2n}^*)| = O(N^{-1/2}) = o(b_n^{-1})$  follows by Holder's inequality. Beginning with  $\text{Var}_*(E_{1n}^*)$  and writing  $E_{1n}^* = |\mathcal{K}_n|^{-1} \sum_{\mathbf{k} \in \mathcal{K}_n} W_{\mathbf{k}}^*$ , it follows from the construction of  $\mathcal{Z}_{n,OL}^*(R_n(\mathbf{k}))$ ,  $\mathbf{k} \in \mathcal{K}_n$ , by the distribution (3) and  $N_1 = b_n^d |\mathcal{K}_n|$  that

$$N_1 \text{Var}_*(E_{1n}^*) = N_1 \frac{\text{Var}_*(W_{\mathbf{0}}^*)}{|\mathcal{K}_n|} = b_n^d \left( \frac{1}{|\mathcal{I}_n|} \sum_{\mathbf{i} \in \mathcal{I}_n} W_{\mathbf{i},cen}^2 - \left( \hat{\nabla}' \hat{\mu}_{n,cen} \right)^2 \right), \quad (11)$$

where  $\hat{\nabla}' \hat{\mu}_{n,cen} = |\mathcal{I}_n|^{-1} \sum_{\mathbf{i} \in \mathcal{I}_n} W_{\mathbf{i},cen}$ . Expand each component  $\hat{c}_\nu$ ,  $\|\nu\|_1 = 1$ , of  $\hat{\nabla}$  around  $\mu$  and write  $W_{\mathbf{i},cen} = S_{1\mathbf{i}} + S_{2\mathbf{i}}$ ,  $\mathbf{i} \in \mathcal{I}_n$ , where  $S_{1\mathbf{i}} = \nabla'(U_{\mathbf{i}} - \mu)$  and  $S_{2\mathbf{i}}$  is defined by the difference. By Assumption 5, Lemma 1(b) and Holder's inequality, we find  $b_n^d E|S_{1\mathbf{i}} S_{2\mathbf{i}}|$ ,  $b_n^d E S_{2\mathbf{i}}^2 \leq \mathcal{C} N^{-1/2}$ ,  $\mathbf{i} \in \mathcal{I}_n$ , and  $b_n^d E(\hat{\nabla}' \hat{\mu}_{n,cen})^2 \leq \mathcal{C} b_n^d N^{-1} = o(b_n^{-1})$  from  $\|\hat{\nabla}\| \leq \mathcal{C}(1 + \|\hat{\mu}_{n,cen}\| + \|\hat{\mu}_{n,cen}\|^{1+a})$ . Hence, (10) for  $\text{Var}_*(E_{1n}^*)$  will follow from (11) by showing

$$\sigma_n^2 - \sigma^2 = O(N^{-1/2}), \quad b_n^d E S_{1\mathbf{i}}^2 - \sigma^2 = -\frac{1}{b_n} \sum_{\mathbf{k} \in \mathbb{Z}^d} \frac{b_n^d - C_n(\mathbf{k})}{b_n^{d-1}} \varrho(\mathbf{k}) = -\frac{B_0}{b_n}(1 + o(1)), \quad (12)$$

using the lattice point count  $C_n(\mathbf{k}) = |\mathbb{Z}^d \cap b_n \mathcal{U} \cap (\mathbf{k} + b_n \mathcal{U})|$ ,  $\mathbf{k} \in \mathbb{Z}^d$ . The first part of (12) holds from Nordman and Lahiri (2004, Lemma 9.1); the second part follows by the Lebesgue dominated convergence theorem since  $0 \leq b_n^d - C_n(\mathbf{k}) \leq \mathcal{C}\|\mathbf{k}\|_\infty^d b_n^{d-1}$ ,  $b_n^{-(d-1)}(b_n^d - C_n(\mathbf{k})) \rightarrow -\|\mathbf{k}\|_\infty$  for  $\mathbf{k} \in \mathbb{Z}^d$  and  $\sum_{\mathbf{k} \in \mathbb{Z}^d} \|\mathbf{k}\|_\infty^d |\varrho(\mathbf{k})| < \infty$ . Turning to  $\text{Var}_*(E_{2n}^*)$  in (10), we bound  $E_*(E_{2n}^*)^2 \leq \mathcal{C}(1 + \|\hat{\mu}_n\|^{2a})(E_*\|\bar{Z}_{n,OL}^* - \hat{\mu}_n\|^4 + E_*\|\bar{Z}_{n,OL}^* - \hat{\mu}_n\|^{4+2a})$ . We next note  $\bar{Z}_{n,OL}^* = |\mathcal{K}_n|^{-1} \sum_{\mathbf{k} \in \mathcal{K}_n} U_{\mathbf{k}}^*$  and, conditioned on the data,  $\{U_{\mathbf{k}}^*\}_{\mathbf{k} \in \mathcal{K}_n}$  are iid with  $E_* U_{\mathbf{0}}^* = \hat{\mu}_n$ ,  $E_* \|U_{\mathbf{0}}^*\|^{4+2a} < \infty$  so that, for  $m = 2$  or  $2 + a$ ,

$$E_* \|\bar{Z}_{n,OL}^* - \hat{\mu}_n\|^{2m} \leq \mathcal{C} |\mathcal{K}_n|^{-m} E_* \|U_{\mathbf{0}}^* - \mu\|^{2m} \leq \mathcal{C} \frac{|\mathcal{K}_n|^{-m}}{|\mathcal{I}_n|} \sum_{\mathbf{i} \in \mathcal{I}_n} \|U_{\mathbf{i}} - \mu\|^{2m}. \quad (13)$$

By Holder's inequality with Lemma 1, we find  $N_1 E \text{Var}_*(E_{2n}^*) \leq N_1 E[E_*(E_{2n}^*)^2] \leq \mathcal{C} N^{-1}$ .  $\square$

**Proof of Theorem 1(ii).** Beginning with (11), write  $|\mathcal{I}_n|^{-1} \sum_{\mathbf{i} \in \mathcal{I}_n} W_{\mathbf{i}}^2 \equiv T_{1n} + 2T_{2n} + T_{3n}$  as the sum of separate averages over terms  $S_{1\mathbf{i}}^2, S_{1\mathbf{i}}S_{2\mathbf{i}}, S_{2\mathbf{i}}^2$ ,  $\mathbf{i} \in \mathcal{I}_n$ , respectively. From Lemma 1(c) and Theorem 9.1 of Nordman and Lahiri (2004), it holds that  $\text{Var}(b_n^d T_{1n}) = 2\sigma^4(2/3)^d b_n^d / \text{vol}(R_n)(1 + o(1))$ . To establish the expression for  $\text{Var}[\hat{\sigma}_{n,OL}^2(b_n)]$ , it now suffices to prove that

$$\text{Var}(N_1 \text{Var}_*(E_{1n}^*)) = \text{Var}(b_n^d T_{1n})(1 + o(1)), \quad \text{Var}(N_1 \text{Var}_*(E_{2n}^*)) = O(N^{-2}), \quad (14)$$

since (10) and (14) entail  $\text{Var}[N_1 \text{Cov}_*(E_{1n}^*, E_{2n}^*)] = O(N^{-1})$ . By (13) and the bound on  $E_*(E_{2n}^*)^2$  preceding (13), we may apply Lemma 1(b) to bound  $\text{Var}[N_1 \text{Var}_*(E_{2n}^*)]$  in (14) by  $N_1^2 E[E_*(E_{2n}^*)^2]^2 \leq \mathcal{C}N^{-2}$ . Then considering  $\text{Var}(N_1 \text{Var}_*(E_{1n}^*))$  in (14), we bound  $\text{Var}[b_n^d (\hat{\nabla}' \hat{\mu}_{n,cen})^2] \leq b_n^{2d} E[\hat{\nabla}' \hat{\mu}_{n,cen}]^4 = O(b_n^{2d} / \text{vol}(R_n)^2)$  so it becomes sufficient from (11) to show  $\text{Var}(b_n^d T_{in}) = o(b_n^d / \text{vol}(R_n))$ ,  $i = 2, 3$ . We consider  $\text{Var}(b_n^d T_{3n})$ ; the proof for  $\text{Var}(b_n^d T_{2n})$  is similar.

For each  $\mathbf{k} \in \mathbb{Z}^d$ , define a covariance function  $\varrho_n(\mathbf{k}) = \text{Cov}(S_{20}^2, S_{2\mathbf{k}}^2)$  and a distance  $\text{dis}_n(\mathbf{k}) = \text{dis}(\mathbf{Z}^d \cap \mathcal{B}_n(\mathbf{0}), \mathbf{Z}^d \cap \mathcal{B}_n(\mathbf{k}))$ . We bound  $\text{Var}(b_n^d T_{3n}) = |\mathcal{I}_n|^{-2} b_n^{2d} \sum_{\mathbf{k} \in \mathbb{Z}^d} |\{\mathbf{i} \in \mathcal{I}_n : \mathbf{i} + \mathbf{k} \in \mathcal{I}_n\}| \cdot \varrho_n(\mathbf{k}) \leq V_{1n} + V_{2n}$  with sums  $V_{1n} = |\mathcal{I}_n|^{-1} b_n^{2d} \sum_{\mathbf{k} \in A_n} |\varrho_n(\mathbf{k})|$ ,  $V_{2n} = |\mathcal{I}_n|^{-1} b_n^{2d} \sum_{\mathbf{k} \in \mathbb{Z}^d \setminus A_n} |\varrho_n(\mathbf{k})|$  where  $A_n = \{\mathbf{k} \in \mathbb{Z}^d : \|\mathbf{k}\|_\infty \leq b_n\}$ . By Lemma 1(b),  $|\varrho_n(\mathbf{k})| \leq E(S_{20}^4) \leq \mathcal{C}N^{-2} b_n^{-2d}$ ,  $\mathbf{k} \in A_n$  holds so that  $V_{1n} = o(b_n^d / \text{vol}(R_n))$ . For  $\mathbf{k} \in \mathbb{Z}^d \setminus A_n$ , it holds that  $\text{dis}_n(\mathbf{k}) \geq 1$  and  $|\varrho_n(\mathbf{k})| \leq \mathcal{C}N^{-2} b_n^{-2d} \min\{1, \alpha_1[\text{dis}_n(\mathbf{k})]^{\delta/(2r+\delta)} b_n^{(2r-1)d-1}\}$  by Lemma 1(a) and Assumption 4<sub>r</sub>. For  $\ell \geq 1 \in \mathbb{Z}_+$ ,  $|\{\mathbf{k} \in \mathbb{Z}^d : \text{dis}_n(\mathbf{k}) = \ell\}| \leq \mathcal{C}(b_n + \ell)^{d-1}$  so we may consider two sums of  $|\varrho_n(\mathbf{k})|$ ,  $\mathbf{k} \in \mathbb{Z}^d \setminus A_n$ , over distances  $\ell \equiv \text{dis}_n(\mathbf{k}) \leq b_n$  or  $> b_n$ :

$$V_{2n} \leq \mathcal{C} \frac{1}{N^2 |\mathcal{I}_n|} \left( \sum_{\ell=1}^{b_n} b_n^{d-1} + b_n^{(2r-1)d-1} \sum_{\ell=b_n+1}^{\infty} \left( \frac{\ell}{b_n} \right)^{2d(r-1)} \ell^{d-1} \alpha_1(\ell)^{\delta/(2r+\delta)} \right) = o(b_n^d / \text{vol}(R_n)),$$

substituting the additional term  $(\ell/b_n)^{2d(r-1)} \geq 1$  in the second sum. Thus,  $\text{Var}(b_n^d T_{3n}) = o(b_n^d / \text{vol}(R_n))$  follows, establishing the claim in (14) for  $\text{Var}_*(E_{1n}^*)$ .  $\square$

**Proof of Theorem 2.** Follows from Theorem 1 and optimization with calculus.  $\square$

**Proof of Theorem 3.** By Theorem 1 and its proof,  $\hat{\sigma}^2$  and  $\hat{B}_0$  are MSE-consistent for  $\sigma^2$  and  $B_0 \neq 0$ , respectively, so the consistency of  $\hat{b}_n^{\text{NPI}}$  then follows.

To establish the consistency of  $\hat{b}_n^{\text{HHJ}}$ , let  $\Delta_{m,\eta}(b) = P(|\widehat{\text{MSE}}_m(b)/\text{MSE}_m(b) - 1| > \eta)$  (with  $m \equiv m_n$ ) for  $\eta > 0$ . It suffices to prove

$$\max_{b \in \mathcal{J}_{m,\epsilon}} \Delta_{m,\eta}(b) \leq \mathcal{C} \left( \frac{m}{\lambda_n} \right)^{d/(d+2)}, \quad \max_{b \in \mathcal{J}_{m,\epsilon}} \left| \left( \frac{b^d C_0}{m^d} + \frac{B_0^2}{b^2} \right)^{-1} \text{MSE}_m(b) - 1 \right| = o(1), \quad (15)$$

where  $\text{MSE}_m(b) = \mathbb{E}[\hat{\sigma}_{\mathbf{0},m}(b) - \sigma^2]^2$  represents a MSE of a size- $b$  block variance estimator on a sampling region  $m\mathcal{U}$  (i.e., spatial observations  $\mathcal{Z}_n(m\mathcal{U})$ ) with  $C_0/[2\sigma^4] = 1$  or  $(2/3)^d$  for NOL or OL block cases. To see this, note  $\hat{b}_m^0$  and  $b_m^{\text{opt}}$  minimize  $\widehat{\text{MSE}}_m(\cdot)$  and  $\text{MSE}_m(\cdot)$ , respectively, where  $b_m^{\text{opt}} = [2m^d B_0^2 / (C_0 d)]^{1/(d+2)}(1 + o(1))$  is the optimal block size for a rectangular region  $m\mathcal{U}$  by Theorem 2. Then, for  $b = \hat{b}_m^0$  or  $b_m^{\text{opt}}$ , it follows that  $\widehat{\text{MSE}}_m(b)/\text{MSE}_m(b) \xrightarrow{p} 1$  from  $\Delta_{m,\eta}(b) \leq \sum_{j \in \mathcal{J}_{m,\epsilon}} \Delta_{m,\eta}(j) = o(1)$  using (15),  $|\mathcal{J}_{m,\epsilon}| = O(m^{d/(d+2)+\epsilon})$  and  $m^\epsilon(m^2/\lambda_n)^{d/(d+2)} = o(1)$ . By this and (15), one may argue that  $\hat{b}_m^0/b_m^{\text{opt}} \xrightarrow{p} 1$ . Consistency of  $\hat{b}_n^{\text{HHJ}} = \hat{b}_m^0 \cdot [\text{vol}(R_n)/m^d]^{1/(d+2)}$  for  $b_n^{\text{opt}}$  then follows.

The second result in (15) for  $\text{MSE}_m(b)$ ,  $b \in \mathcal{J}_{m,\epsilon}$ , follows from the proof of Theorem 1 and using  $\text{vol}(m\mathcal{U}) = m^d$ ; note  $\sigma^2$  may be used in Theorem 1(i) (in place of  $\sigma_n^2$ ). For large  $n$ , note that  $\mathbb{E}[\hat{\sigma}_n^2(b_n^*) - \sigma^2]^2 \leq \mathcal{C}\lambda_n^{-2d/(d+2)}$  (since  $b_n^*$  is of order  $\lambda_n^{d/(d+2)}$ ) by applying Theorem 1 and that  $\text{MSE}_m(b) \geq \mathcal{C}m^{-2d/(d+2)}$  (the optimal MSE order) for  $b \in \mathcal{J}_{m,\epsilon}$ . Define  $\overline{\text{MSE}}_m(b)$  by replacing  $\hat{\sigma}_n^2(b_n^*)$  with  $\sigma^2$  in the definition of  $\widehat{\text{MSE}}_m(b)$  from (6). By adding/subtracting  $\sigma^2$  inside  $\widehat{\text{MSE}}_m(b)$  and applying the Markov inequality, it holds that  $\Delta_{m,\eta}(b) \leq \mathcal{C}[P_1(b) + P_2(b) + P_3(b)]$  for any  $b \in \mathcal{J}_{m,\epsilon}$ , where  $P_1(b) = \{\text{Var}[\overline{\text{MSE}}_m(b)]/\text{MSE}_m^2(b)\}^{1/2}$ ,  $P_2(b) = \mathbb{E}[\hat{\sigma}_n^2(b_n^*) - \sigma^2]^2/\text{MSE}_m(b) \leq \mathcal{C}(m/\lambda_n)^{2d/(d+2)}$  and by Holder's inequality  $P_3(b) = \sum_{\mathbf{i} \in \mathcal{M}_m} \mathbb{E}[\hat{\sigma}_n^2(b_n^*) - \sigma^2]|\hat{\sigma}_{\mathbf{i},m}^2(b) - \sigma^2|/(\text{MSE}_m(b)|\mathcal{M}_n|) \leq \mathcal{C}P_2^{1/2}(b) = O((m/\lambda_n)^{d/(d+2)})$ . To handle  $P_1(b)$ , arguments as in proof of Theorem 1(ii) yield  $\text{Var}[\overline{\text{MSE}}_m(b)] \leq \mathcal{C}\lambda_n^{-d}m^d\mathbb{E}[\hat{\sigma}_{\mathbf{0},m}^2(b) - \sigma^2]^4$ ,  $b \in \mathcal{J}_{m,\epsilon}$ . The bias/variance arguments from the proof of Theorem 1 give  $\mathbb{E}[\hat{\sigma}_{\mathbf{0},m}^2(b) - \sigma^2]^4/\text{MSE}_m^2(b) \leq \mathcal{C}$ , for any  $b \in \mathcal{J}_{m,\epsilon}$  when  $n$  is large, using the second part of (15). Hence,  $P_1(b) \leq \mathcal{C}(m/\lambda_n)^{d/2}$  holds and the bound on  $\max_{b \in \mathcal{J}_{m,\epsilon}} \Delta_{m,\eta}(b)$  in (15) follows.  $\square$

## References

- BESAG, J. (1974) Spatial interaction and the statistical analysis of lattice systems (with discussion). *J. R. Stat. Soc. Ser. B* **36**, 192-236.
- BÜHLMANN, P. and KÜNSCH, H. R. (1999) Block length selection in the bootstrap for time series. *Comput. Stat. Data Anal.* **31**, 295-310.
- CHAN G. and WOOD A. T. A. (1997) An algorithm for simulating stationary Gaussian random fields. *Applied Statistics* **46**, 171-181.
- CRESSIE, N. (1993) *Statistics for Spatial Data*, 2nd Edition. John Wiley & Sons, New York.
- DAVID, F. N. and MOORE, P. G. (1954) Notes on contagious distributions in plant populations. *Ann. Bot.* **18**, 47-53.

- DOUKHAN, P. (1994) Mixing: properties and examples. *Lecture Notes in Statistics* **85**. Springer-Verlag, New York.
- GUYON, X. (1995) *Random Fields on a Network*. Springer-Verlag, New York.
- HALL, P. (1985) Resampling a coverage pattern. *Stochastic Processes and Their Applications* **20**, 231-246.
- HALL, P. (1992) *The Bootstrap and Edgeworth Expansion*. Springer-Verlag, New York.
- HALL, P. and JING, B.-Y. (1996) On sample reuse methods for dependent data. *J. R. Stat. Soc. Ser. B* **58**, 727-737.
- HALL, P., HOROWITZ, J. L., and JING, B.-Y. (1995) On blocking rules for the bootstrap with dependent data. *Biometrika* **82**, 561-574.
- KÜNSCH, H. R. (1989) The jackknife and the bootstrap for general stationary observations. *Ann. Statist.* **17**, 1217-1241.
- LAHIRI, S. N. (1996) Empirical choice of the optimal block length for block bootstrap methods. Preprint, Department of Statistics, Iowa State University, Ames, IA.
- LAHIRI, S. N. (1999) Theoretical comparisons of block bootstrap methods. *Ann. Statist.* **27**, 386-404.
- LAHIRI, S. N. (2003a) *Resampling Methods for Dependent Data*. Springer, New York.
- LAHIRI, S. N. (2003b) Central limit theorems for weighted sums of a spatial process under a class of stochastic and fixed designs. *Sankhya: Series A* **65**, 356-388.
- LAHIRI, S. N., FURUKAWA, K. and LEE, Y.-D. (2006) A nonparametric plug-in rule for selecting optimal block lengths for block bootstrap methods. *Stat. Methodol.* (in press)
- LIU, R.Y. and SINGH, K. (1992) Moving blocks jackknife and bootstrap capture weak dependence. In *Exploring the Limits of Bootstrap*, 225-248, R. LePage and L. Billard (editors). John Wiley & Sons, New York.
- NORDMAN, D. J. and LAHIRI, S. N. (2004) On optimal spatial subsample size for variance estimation. *Ann. Statist.* **32**, 1981-2027.
- POLITIS, D. N. and ROMANO, J. P. (1993) Nonparametric resampling for homogeneous strong mixing random fields. *J. Multivariate Anal.* **47**, 301-328.
- POLITIS, D. N. and ROMANO, J. P. (1994a) Large sample confidence regions based on subsamples under minimal assumptions. *Ann. Statist.* **22**, 2031-2050.
- POLITIS, D. N. and ROMANO, J. P. (1994b) The stationary bootstrap. *Journal of the American Statistical Association* **89**, 1303-1313.

- POLITIS, D. N. and SHERMAN, M. (2001) Moment estimation for statistics from marked point processes. *J. R. Stat. Soc. Ser. B* **63**, 261-275.
- POLITIS, D. N., PAPARODITIS, E. and ROMANO, J. P. (1999). Resampling marked point processes. In *Multivariate Analysis, Design of Experiments, and Survey Sampling: a Tribute to J. N. Srivastava* (Ed. - S Ghosh). Marcel Dekker, New York, 163-185.
- POLITIS, D. N., ROMANO, J. P., and WOLF, M. (1999) *Subsampling*. Springer, New York.
- POLITIS, D. N. and WHITE, H. (2004) Automatic block-length selection for the dependent bootstrap. *Econometric Reviews* **23** 53-70.
- SHERMAN, M. (1996) Variance estimation for statistics computed from spatial lattice data. *J. R. Stat. Soc. Ser. B* **58**, 509-523.
- SHERMAN, M. and CARLSTEIN, E. (1994) Nonparametric estimation of the moments of a general statistic computed from spatial data. *J. Amer. Statist. Assoc.* **89**, 496-500.
- ZHU, J. and LAHIRI, S.N. (2007) Weak convergence of blockwise bootstrapped empirical processes for stationary random fields with statistical applications. *Stat. Inference Stoch. Process.* **10**, 107-145.

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Figure 1: The blocking mechanism for the spatial block bootstrap method. (a) Partition of a hexagonal sampling region  $R_n$  by subregions  $R_n(\mathbf{k})$ ,  $\mathbf{k} \in \mathcal{K}_n$  from (2); (b) Set of OL complete blocks; (c) Set of NOL complete blocks; (d) Bootstrap version of the spatial process  $Z(\cdot)$  obtained by resampling and concatenating blocks from (b) or (c) in positions of complete blocks from (a).

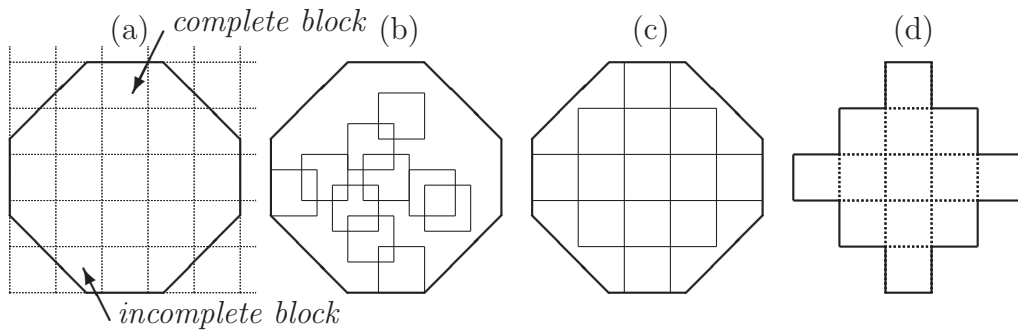


Table 1: Normalized bias  $E\hat{\sigma}_n^2/\sigma_n^2 - 1$  and variance  $\text{Var}(\hat{\sigma}_n^2)/\sigma_n^4$  for OL block bootstrap and OL subsampling estimators of  $\sigma_n^2 = N\text{Var}(\bar{Z}_N)$  when using optimal block scaling from Figure 7). Estimates based on 100,000 simulations.

|                    | $R_n = \{\mathbf{x} \in \mathbb{R}^2 : \ \mathbf{x}\  \leq 9\}$ |        |                |        | $R_n = \{\mathbf{x} \in \mathbb{R}^2 : \ \mathbf{x}\  \leq 20\}$ |        |                |        |
|--------------------|-----------------------------------------------------------------|--------|----------------|--------|------------------------------------------------------------------|--------|----------------|--------|
|                    | Model(1,1)                                                      |        | Model(0.5,0.3) |        | Model(1,1)                                                       |        | Model(0.5,0.3) |        |
|                    | Bias                                                            | Var    | Bias           | Var    | Bias                                                             | Var    | Bias           | Var    |
| <i>Bootstrap</i>   | -0.3895                                                         | 0.0478 | -0.6958        | 0.0283 | -0.2629                                                          | 0.0263 | -0.4814        | 0.0543 |
| <i>Subsampling</i> | -0.3855                                                         | 0.0671 | -0.7003        | 0.0328 | -0.2679                                                          | 0.0283 | -0.4754        | 0.0536 |

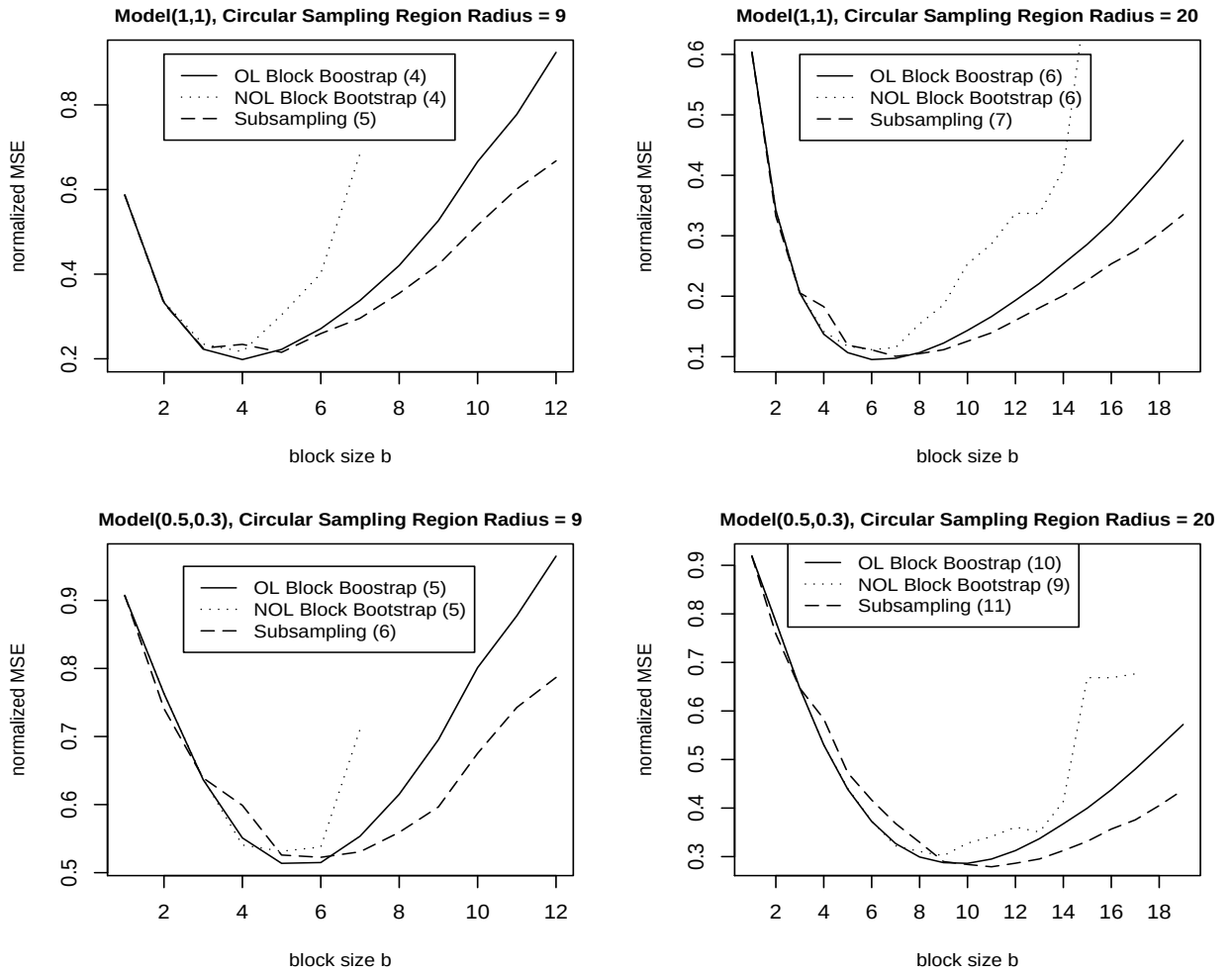


Figure 2: Estimates of normalized MSE,  $E(\hat{\sigma}_n^2/\sigma_n^2 - 1)^2$ , for OL/NOL spatial block bootstrap variance estimators  $\hat{\sigma}_n^2(b_n)$  of  $\sigma_n^2 = N\text{Var}(\bar{Z}_N)$  for various block sizes  $b_n$  as well as OL subsampling variance estimator. Each estimated MSE is based on 100,000 simulations. Values in  $(\cdot)$  denote the optimal block sizes with minimal MSE.



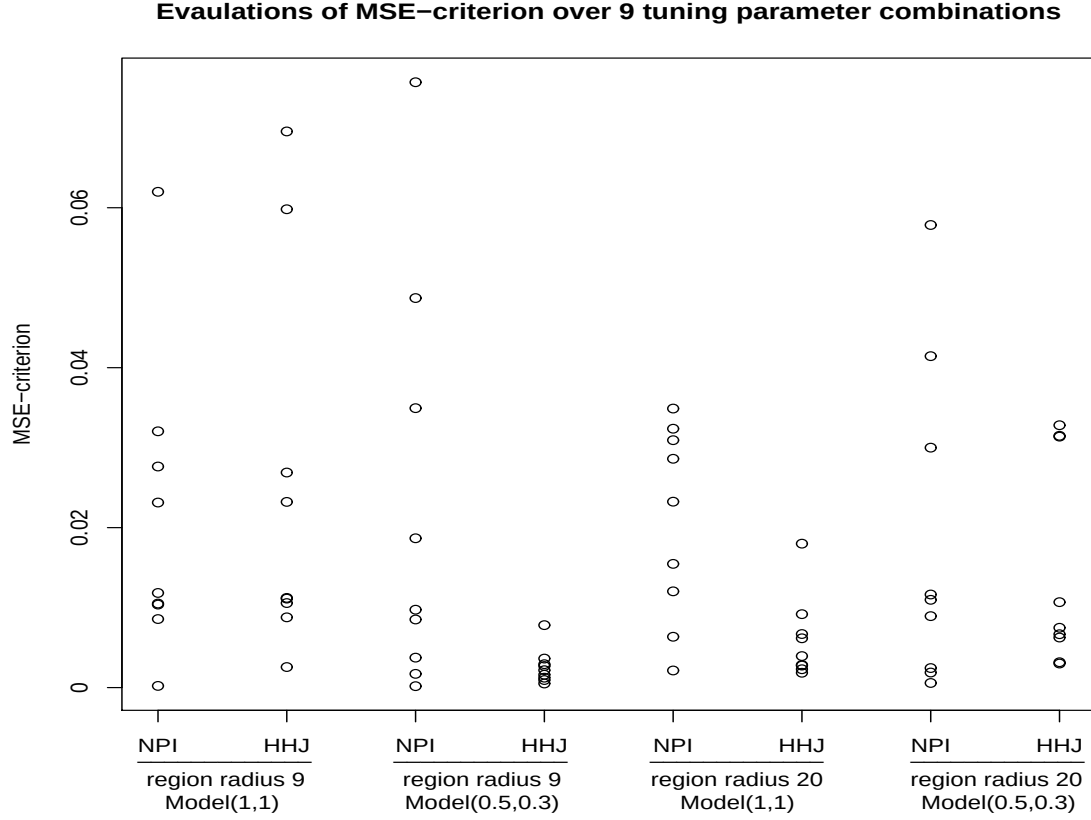


Figure 3: Dot plots of MSE-criterion (8) based on 9 combinations of tuning parameter coefficients for each block selection method, circular sampling region  $R_n$  and covariance model. The NPI method uses  $C_1, C_2 \in \{0.5, 0.75, 1\}$ ; the HHJ method uses  $C^* \in \{1, 1.5, 2\}$ ,  $m \in \{C \text{vol}(R_n)^{1/6} : C = 2, 3, 4\}$ .

Table 2: Frequency distribution of empirically chosen OL block bootstrap scaling with NPI and HHJ methods (based on 1000 simulations). Along with  $C_2 = 0.5$ , NPI1 and NPI2 use  $C_1 = 0.5$  and  $0.75$ , respectively. With  $C^* = 1$ , the HHJ  $\equiv$  HHJ $_m$  method uses  $m = 5, 10$  and  $15$ , respectively. True optimal block sizes  $b_{n,OL}^{\text{opt}}$  are determined from Figure 2.

| $R_n$ radius,<br>$Model(\cdot, \cdot)$ |             | Method |     |     |     |     |     |     |     |     |     |     | $b_{n,OL}^{\text{opt}}$ |
|----------------------------------------|-------------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------------------------|
|                                        |             | 2      | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  |                         |
| 9,<br>(1,1)                            | NPI1        |        | 76  | 924 |     |     |     |     |     |     |     |     | 4                       |
|                                        | NPI2        |        | 646 | 354 |     |     |     |     |     |     |     |     |                         |
|                                        | HHJ $_5$    | 103    |     | 897 |     |     |     |     |     |     |     |     |                         |
|                                        | HHJ $_{10}$ | 8      | 288 | 597 | 104 | 2   |     | 1   |     |     |     |     |                         |
| 20,<br>(1,1)                           | NPI1        |        |     |     | 8   | 78  | 284 | 458 | 162 | 10  |     |     | 6                       |
|                                        | NPI2        |        |     |     | 11  | 127 | 558 | 299 | 5   |     |     |     |                         |
|                                        | HHJ $_{10}$ |        |     | 119 |     | 871 |     | 10  |     |     |     |     |                         |
|                                        | HHJ $_{15}$ |        | 22  |     | 427 | 481 |     | 69  | 1   |     |     |     |                         |
| 9,<br>(0.5,0.3)                        | NPI1        |        |     | 331 | 669 |     |     |     |     |     |     |     | 5                       |
|                                        | NPI2        |        | 145 | 855 |     |     |     |     |     |     |     |     |                         |
|                                        | HHJ $_5$    |        |     | 997 | 3   |     |     |     |     |     |     |     |                         |
|                                        | HHJ $_{10}$ |        | 19  | 441 | 477 | 58  |     | 4   | 1   |     |     |     |                         |
| 20,<br>(0.5,0.3)                       | NPI1        |        |     |     |     |     |     | 1   | 9   | 200 | 624 | 166 | 10                      |
|                                        | NPI2        |        |     |     |     |     |     | 6   | 744 | 250 |     |     |                         |
|                                        | HHJ $_{10}$ |        |     |     |     | 6   |     | 926 |     |     |     |     |                         |
|                                        | HHJ $_{15}$ |        |     |     | 5   | 242 |     | 657 | 95  |     | 1   |     |                         |

Table 3: Normalized MSEs  $E(\hat{\sigma}_{n,OL}^2/\sigma_n^2 - 1)^2$  for OL block bootstrap estimator  $\hat{\sigma}_{n,OL}^2 \equiv \hat{\sigma}_{n,OL}^2(b_n)$  of MPLE variance  $\sigma_n^2 = N\text{Var}(\hat{\beta}_n)$ , based on 1000 simulations for each block  $b_n$ . Also included, the distribution of OL block size estimates  $\hat{b}_{n,OL}^{\text{NPI}}$  using NPI method with  $C_1 = C_2 = 0.5$  (i.e., percentage of times that  $\hat{b}_{n,OL}^{\text{NPI}} = b_n$  based on additional 500 simulations).

| block size $b_n$                      | 1     | 2     | 3      | 4     | 5     | 6     | 7     |
|---------------------------------------|-------|-------|--------|-------|-------|-------|-------|
| MSE $\times 10^2$                     | 1.654 | 0.939 | 0.896* | 1.006 | 1.326 | 1.879 | 2.450 |
| $\hat{b}_{n,OL}^{\text{NPI}} = b_n$ % | 9.0%  | 16.4% | 24.8%  | 23.8% | 16.0% | 8.2%  | 1.8%  |